

Data Analysis Comparison Between SQL and Python

1. SQL Query: select * from dataset_1;

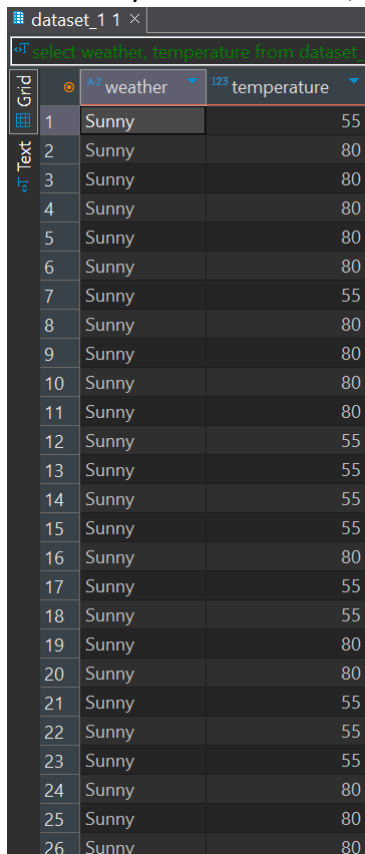
	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	has_children	education	occupation	income
1	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
3	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
5	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
6	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
7	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
10	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
11	No Urgent Place	Kid(s)	Sunny	80	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
12	No Urgent Place	Kid(s)	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
13	No Urgent Place	Kid(s)	Sunny	55	6PM	Coffee House	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
14	Home	Alone	Sunny	55	6PM	Bar	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
15	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
16	Home	Alone	Sunny	80	6PM	Coffee House	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
17	Work	Alone	Sunny	55	7AM	Coffee House	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
18	Work	Alone	Sunny	55	7AM	Bar	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
19	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
20	Work	Alone	Sunny	80	7AM	Carry out & Take away	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
21	Work	Alone	Sunny	55	7AM	Restaurant(<20)	1d	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
22	Work	Alone	Sunny	55	7AM	Coffee House	2h	Female	21	Unmarried partner	1	Some college - no degree	Unemployed	\$37500 - \$49999
23	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999
24	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999
25	No Urgent Place	Friend(s)	Sunny	80	10AM	Bar	1d	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999
26	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999
27	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999
28	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999
29	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999
30	No Urgent Place	Friend(s)	Sunny	80	2PM	Restaurant(<20)	1d	Male	21	Single	0	Bachelors degree	Architecture & Engineering	\$62500 - \$74999

Python Code: df

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway	RestaurantLessThan20	Restaurant20To50
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Male	26	Single	...	1~3	4~8	1~3
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Male	26	Single	...	1~3	4~8	1~3
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Male	26	Single	...	1~3	4~8	1~3
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Male	26	Single	...	1~3	4~8	1~3
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Male	26	Single	...	1~3	4~8	1~3

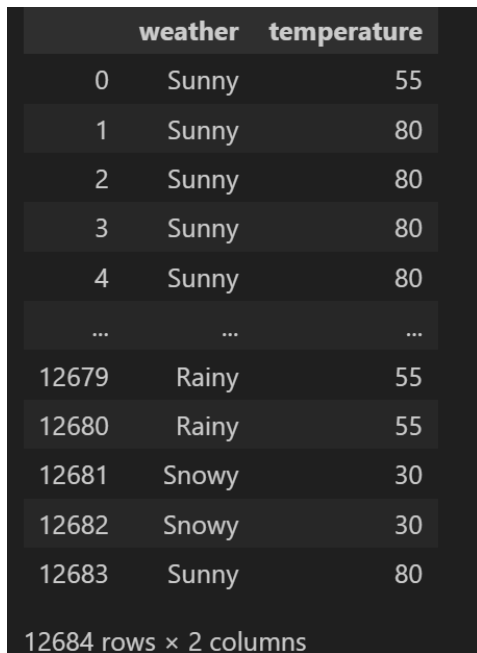
12684 rows x 27 columns

2. SQL Query: select weather, temperature from dataset_1;



	weather	temperature
1	Sunny	55
2	Sunny	80
3	Sunny	80
4	Sunny	80
5	Sunny	80
6	Sunny	80
7	Sunny	55
8	Sunny	80
9	Sunny	80
10	Sunny	80
11	Sunny	80
12	Sunny	55
13	Sunny	55
14	Sunny	55
15	Sunny	55
16	Sunny	80
17	Sunny	55
18	Sunny	55
19	Sunny	80
20	Sunny	80
21	Sunny	55
22	Sunny	55
23	Sunny	55
24	Sunny	80
25	Sunny	80
26	Sunny	80

Python Code: df[['weather', 'temperature']]



	weather	temperature
0	Sunny	55
1	Sunny	80
2	Sunny	80
3	Sunny	80
4	Sunny	80
...	...	...
12679	Rainy	55
12680	Rainy	55
12681	Snowy	30
12682	Snowy	30
12683	Sunny	80

12684 rows × 2 columns

3. SQL Query: select * from dataset_1 limit 5;

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	has_children	education	occupation	income	car
1	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999	
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999	
3	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999	
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999	
5	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999	

Python Code: df.head()

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway	RestaurantLessThan20	Restaurant20To50
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3

5 rows × 27 columns

4. SQL Query: select distinct passanger from dataset_1;

	passanger
1	Alone
2	Friend(s)
3	Kid(s)
4	Partner

Python Code: df['passanger'].unique()

```
array(['Alone', 'Friend(s)', 'Kid(s)', 'Partner'], dtype=object)
```

5. SQL Query: select * from dataset_1 where destination = 'Home';

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	has_children	education	occupation	income
1	Home	Alone	Sunny	55	6PM	Bar	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
2	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
3	Home	Alone	Sunny	80	6PM	Coffee House	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
4	Home	Alone	Sunny	55	6PM	Bar	1d	Male	21	Single		0 Bachelors degree	Architecture & Engineering	\$62500 - \$74999
5	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Male	21	Single		0 Bachelors degree	Architecture & Engineering	\$62500 - \$74999
6	Home	Alone	Sunny	80	6PM	Coffee House	2h	Male	21	Single		0 Bachelors degree	Architecture & Engineering	\$62500 - \$74999
7	Home	Alone	Sunny	55	6PM	Bar	1d	Male	46	Single		0 Some college - no degree	Student	\$12500 - \$24999
8	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Male	46	Single		0 Some college - no degree	Student	\$12500 - \$24999
9	Home	Alone	Sunny	80	6PM	Coffee House	2h	Male	46	Single		0 Some college - no degree	Student	\$12500 - \$24999

Python Code: df[df['destination']=='Home']

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway	RestaurantLessThan20	Restaurant20To50
13	Home	Alone	Sunny	55	6PM	Bar	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3
14	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3
15	Home	Alone	Sunny	80	6PM	Coffee House	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
35	Home	Alone	Sunny	55	6PM	Bar	1d	Male	21	Single	...	4~8	4~8	less1
36	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Male	21	Single	...	4~8	4~8	less1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
12675	Home	Alone	Snowy	30	10PM	Coffee House	2h	Male	26	Single	...	1~3	4~8	1~3
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	Male	26	Single	...	1~3	4~8	1~3
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	Male	26	Single	...	1~3	4~8	1~3
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	Male	26	Single	...	1~3	4~8	1~3
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Male	26	Single	...	1~3	4~8	1~3

3237 rows × 27 columns

6. SQL Query: select * from dataset_1 order by coupon;

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	has_children	education	occupation
1	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed
2	Home	Alone	Sunny	55	6PM	Bar	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed
3	Work	Alone	Sunny	55	7AM	Bar	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed
4	No Urgent Place	Friend(s)	Sunny	80	10AM	Bar	1d	Male	21	Single		0 Bachelors degree	Architecture & Engineering
5	Home	Alone	Sunny	55	6PM	Bar	1d	Male	21	Single		0 Bachelors degree	Architecture & Engineering
6	Work	Alone	Sunny	55	7AM	Bar	1d	Male	21	Single		0 Bachelors degree	Architecture & Engineering
7	No Urgent Place	Friend(s)	Sunny	80	10AM	Bar	1d	Male	46	Single		0 Some college - no degree	Student
8	Home	Alone	Sunny	55	6PM	Bar	1d	Male	46	Single		0 Some college - no degree	Student
9	Work	Alone	Sunny	55	7AM	Bar	1d	Male	46	Single		0 Some college - no degree	Student
10	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Male	46	Married partner		1 Bachelors degree	Education&Training&Library
11	Home	Alone	Sunny	55	6PM	Bar	1d	Male	46	Married partner		1 Bachelors degree	Education&Training&Library
12	Work	Alone	Sunny	55	7AM	Bar	1d	Male	46	Married partner		1 Bachelors degree	Education&Training&Library
13	No Urgent Place	Friend(s)	Sunny	80	10AM	Bar	1d	Male	21	Single		0 Associates degree	Student
14	Home	Alone	Sunny	55	6PM	Bar	1d	Male	21	Single		0 Associates degree	Student

Python Code: df.sort_values('coupon

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway	RestaurantLessThan20	Restaurant20To50
11702	Home	Partner	Sunny	30	10PM	Bar	2h	Female	50plus	Married partner	...	4~8	1~3	less1
9930	No Urgent Place	Alone	Snowy	30	2PM	Bar	1d	Female	21	Single	...	gt8	gt8	4~8
10632	Home	Alone	Rainy	55	6PM	Bar	1d	Male	21	Single	...	gt8	less1	less1
7997	No Urgent Place	Friend(s)	Rainy	55	10PM	Bar	2h	Male	26	Unmarried partner	...	4~8	never	1~3
11166	Work	Alone	Snowy	30	7AM	Bar	1d	Female	41	Married partner	...	gt8	1~3	less1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
10476	Home	Alone	Sunny	80	6PM	Restaurant(<20)	1d	Female	31	Unmarried partner	...	1~3	1~3	less1
5447	Home	Alone	Sunny	80	10PM	Restaurant(<20)	2h	Female	50plus	Single	...	less1	less1	never
10478	Home	Alone	Snowy	30	10PM	Restaurant(<20)	2h	Female	31	Unmarried partner	...	1~3	1~3	less1
5440	No Urgent Place	Alone	Sunny	80	2PM	Restaurant(<20)	2h	Female	50plus	Single	...	less1	less1	never
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3

12684 rows × 27 columns

7. SQL Query: select destination as DESTINATION from dataset_1;

select destination as DESTINATION

Grid		A-Z DESTINATION
	1	No Urgent Place
Text	2	No Urgent Place
	3	No Urgent Place
	4	No Urgent Place
	5	No Urgent Place
	6	No Urgent Place
	7	No Urgent Place

Python Code: `df.rename(columns={'destination':'DESTINATION'}, inplace=True)`
df

	DESTINATION	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway	RestaurantLessThan20	Restaurant20To50
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	...	NaN	4~8	1~3
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	...	NaN	4~8	1~3
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Male	26	Single	...	1~3	4~8	1~3
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Male	26	Single	...	1~3	4~8	1~3
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Male	26	Single	...	1~3	4~8	1~3
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Male	26	Single	...	1~3	4~8	1~3
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Male	26	Single	...	1~3	4~8	1~3

12684 rows x 27 columns

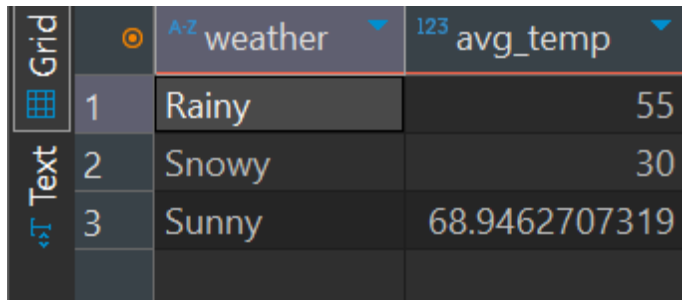
8. SQL Query: select occupation from dataset_1 group by occupation;

Grid	⊙	A-Z occupation
Text	1	Architecture & Engineering
	2	Arts Design Entertainment Sports & Media
	3	Building & Grounds Cleaning & Maintenance
	4	Business & Financial
	5	Community & Social Services
	6	Computer & Mathematical
	7	Construction & Extraction
	8	Education&Training&Library
	9	Farming Fishing & Forestry
	10	Food Preparation & Serving Related
	11	Healthcare Practitioners & Technical
	12	Healthcare Support
	13	Installation Maintenance & Repair
	14	Legal
	15	Life Physical Social Science
	16	Management
	17	Office & Administrative Support
	18	Personal Care & Service
	19	Production Occupations
	20	Protective Service
	21	Retired
	22	Sales & Related
	23	Student
Record	24	Transportation & Material Moving
	25	Unemployed

Python Code: df.groupby('occupation').size().to_frame('Count').reset_index()

	occupation	Count
0	Architecture & Engineering	175
1	Arts Design Entertainment Sports & Media	629
2	Building & Grounds Cleaning & Maintenance	44
3	Business & Financial	544
4	Community & Social Services	241
5	Computer & Mathematical	1408
6	Construction & Extraction	154
7	Education&Training&Library	943
8	Farming Fishing & Forestry	43
9	Food Preparation & Serving Related	298
10	Healthcare Practitioners & Technical	244
11	Healthcare Support	242
12	Installation Maintenance & Repair	133
13	Legal	219
14	Life Physical Social Science	170
15	Management	838
16	Office & Administrative Support	639
17	Personal Care & Service	175
18	Production Occupations	110
19	Protective Service	175
20	Retired	495
21	Sales & Related	1093
22	Student	1584
23	Transportation & Material Moving	218
24	Unemployed	1870

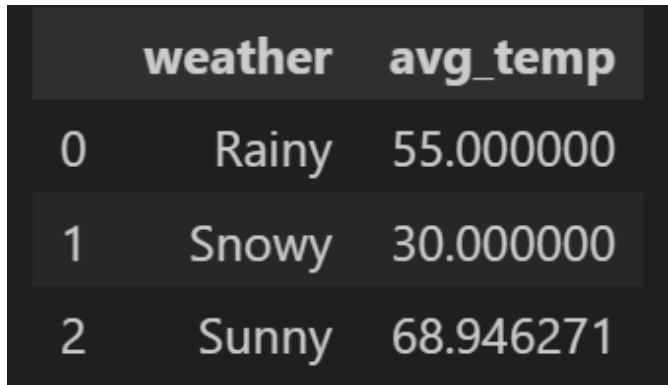
9. SQL Query: select weather, AVG(temperature) as avg_weather from dataset_1 group by weather;



A screenshot of a data table interface. The table has two columns: 'weather' and 'avg_temp'. The 'weather' column has three rows: 'Rainy', 'Snowy', and 'Sunny'. The 'avg_temp' column has corresponding values: 55, 30, and 68.9462707319. The interface includes a 'Grid' view icon and a 'Text' view icon.

	weather	avg_temp
1	Rainy	55
2	Snowy	30
3	Sunny	68.9462707319

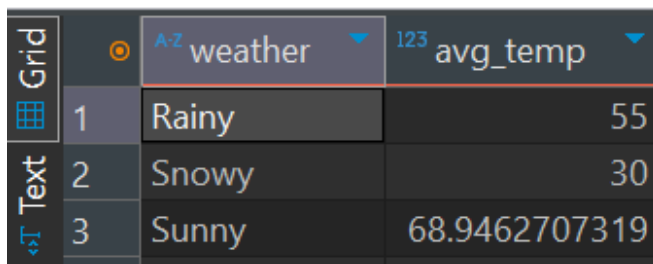
Python Code: `df.groupby('weather')['temperature'].mean().to_frame('avg_temp').reset_index()`



A screenshot of a data table interface. The table has two columns: 'weather' and 'avg_temp'. The 'weather' column has three rows: 'Rainy', 'Snowy', and 'Sunny'. The 'avg_temp' column has corresponding values: 55.000000, 30.000000, and 68.946271. The interface includes a 'Grid' view icon and a 'Text' view icon.

	weather	avg_temp
0	Rainy	55.000000
1	Snowy	30.000000
2	Sunny	68.946271

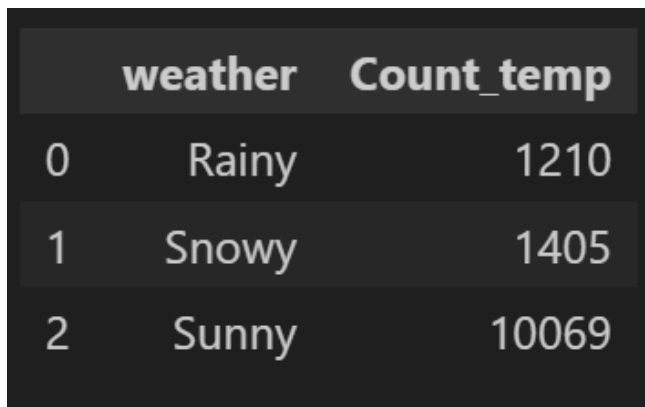
10. SQL Query: select weather, COUNT(temperature) as count_temp group by weather;



A screenshot of a data table interface. The table has two columns: 'weather' and 'avg_temp'. The 'weather' column has three rows: 'Rainy', 'Snowy', and 'Sunny'. The 'avg_temp' column has corresponding values: 55, 30, and 68.9462707319. The interface includes a 'Grid' view icon and a 'Text' view icon.

	weather	avg_temp
1	Rainy	55
2	Snowy	30
3	Sunny	68.9462707319

Python Code: `df.groupby('weather')['temperature'].size().to_frame('Count_temp').reset_index()`



A screenshot of a data table interface. The table has two columns: 'weather' and 'Count_temp'. The 'weather' column has three rows: 'Rainy', 'Snowy', and 'Sunny'. The 'Count_temp' column has corresponding values: 1210, 1405, and 10069. The interface includes a 'Grid' view icon and a 'Text' view icon.

	weather	Count_temp
0	Rainy	1210
1	Snowy	1405
2	Sunny	10069

11. SQL Query: select weather, COUNT(DISTINCT temperature) as count_dist_temp from dataset_1 group by weather;

Grid		A-Z weather	123 count_dist_temp
1		Rainy	1
2		Snowy	1
3		Sunny	3

Python Code:

```
df.groupby('weather')['temperature'].nunique().to_frame('count_dist_temp').reset_index()
```

	weather	count_dist_temp
0	Rainy	1
1	Snowy	1
2	Sunny	3

12. SQL Query: select weather, SUM(temperature) as temp_sum from dataset_1 group by weather;

Grid		A-Z weather	123 temp_sum
1		Rainy	66,550
2		Snowy	42,150
3		Sunny	694,220

Python Code: df.groupby('weather')['temperature'].sum().to_frame('sum_temp').reset_index()

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

13. SQL Query: select weather, MIN(temperature) as *min_temp* from dataset_1 group by weather;

Grid		A-Z weather	123 min_temp
1		Rainy	55
2		Snowy	30
3		Sunny	30

Python Code: `df.groupby('weather')['temperature'].min().to_frame('min_temp').reset_index()`

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	30

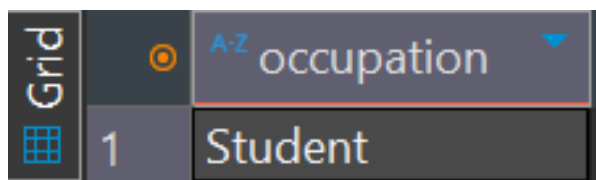
14. SQL Query: select weather, MAX(temperature) as *max_temp* from dataset_1 group by weather;

Grid		A-Z weather	123 max_temp
1		Rainy	55
2		Snowy	30
3		Sunny	80

Python Code: `df.groupby('weather')['temperature'].max().to_frame('max_temp').reset_index()`

	weather	max_temp
0	Rainy	55
1	Snowy	30
2	Sunny	80

15. SQL Query: select occupation from dataset_1 group by occupation having occupation='Student';

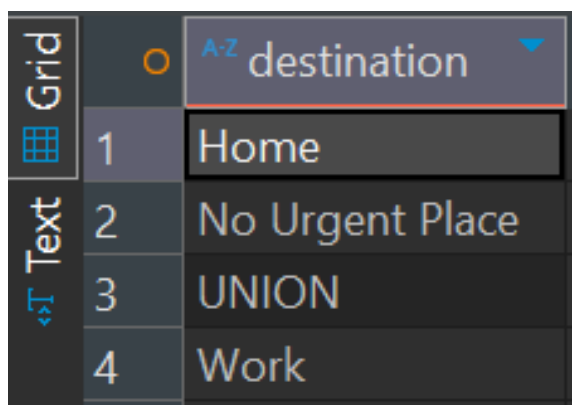


Grid	1	occupation
		Student

Python Code: `df.groupby('occupation').filter(lambda x: x['occupation'].iloc[0] == 'Student').groupby('occupation').size()`

```
occupation
Student    1584
dtype: int64
```

16. SQL Query: select DISTINCT destination from(select * from dataset_1 union select * from table_to_union);



Grid	1	destination
		Home
Text	2	No Urgent Place
	3	UNION
	4	Work

Python Code: `pd.concat([df, df])['DESTINATION'].drop_duplicates()`

```
0    No Urgent Place
13           Home
16           Work
Name: DESTINATION, dtype: object
```

17. SQL Query: select *a*.destination,*a*.time,*b*.part_of_day from dataset_1 *a* inner join table_to_join *b* on *a*.time=*b*.time;

Grid		A-Z destination	A-Z time	A-Z part_of_day
Text	1	No Urgent Place	2PM	Afternoon
	2	No Urgent Place	10AM	Morning
	3	No Urgent Place	10AM	Morning
	4	No Urgent Place	2PM	Afternoon
	5	No Urgent Place	2PM	Afternoon
	6	No Urgent Place	6PM	Evening
	7	No Urgent Place	2PM	Afternoon
	8	No Urgent Place	10AM	Morning
	9	No Urgent Place	10AM	Morning
	10	No Urgent Place	10AM	Morning
	11	No Urgent Place	2PM	Afternoon
	12	No Urgent Place	2PM	Afternoon
	13	No Urgent Place	6PM	Evening
Record	14	Home	6PM	Evening
	15	Home	6PM	Evening
	16	Home	6PM	Evening
	17	Work	7AM	Morning

Python Code: `pd.merge(df, df[['time', 'part_of_day']], on='time', how='inner')[['destination', 'time', 'part_of_day']]`

18. SQL Query: select destination, passanger FROM (select * from dataset_1 where passanger='Alone');

Grid		A-Z destination	A-Z passanger
1		No Urgent Place	Alone
2		Home	Alone
3		Home	Alone
4		Home	Alone
5		Work	Alone
6		Work	Alone
7		Work	Alone
8		Work	Alone
9		Work	Alone
10		Work	Alone
11		No Urgent Place	Alone
12		No Urgent Place	Alone
13		Home	Alone
14		Home	Alone
15		Home	Alone

Python Code: df[df['passanger'] == 'Alone'][['DESTINATION', 'passanger']]

	DESTINATION	passanger
0	No Urgent Place	Alone
13	Home	Alone
14	Home	Alone
15	Home	Alone
16	Work	Alone
...	...	...
12676	Home	Alone
12680	Work	Alone
12681	Work	Alone
12682	Work	Alone
12683	Work	Alone

7305 rows × 2 columns

19. SQL Query: select * from dataset_1 where weather like 'Sun%';

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	has_children	education	occupation	income
1	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
3	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
5	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
6	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
7	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
10	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
11	No Urgent Place	Kid(s)	Sunny	80	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999
12	No Urgent Place	Kid(s)	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner		1 Some college - no degree	Unemployed	\$37500 - \$49999

Python Code: df[df['weather'].str.startswith('Sun')]

	DESTINATION	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	NaN
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	...	NaN
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	NaN
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	...	NaN
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	...	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
12673	Home	Alone	Sunny	30	6PM	Carry out & Take away	1d	Male	26	Single	...	1~3
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	Male	26	Single	...	1~3
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	Male	26	Single	...	1~3
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	Male	26	Single	...	1~3
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Male	26	Single	...	1~3

10069 rows × 27 columns

20. SQL Query: select DISTINCT temperature from dataset_1 where temperature between 29 and 75;

Grid		123 temperature
1		55
Text	2	30

Python Code: df[(df['temperature']<=29) & (df['temperature']<=75)]['temperature'].unique()

```
array([], dtype=int64)
```

21. SQL Query: select occupation from dataset_1 where occupation in('Sales & Related','Management');

Grid		A-Z occupation
Text	1	Sales & Related
	2	Sales & Related
	3	Sales & Related
	4	Sales & Related
	5	Sales & Related
	6	Sales & Related
	7	Sales & Related
	8	Sales & Related
	9	Sales & Related
	10	Sales & Related
	11	Sales & Related
	12	Sales & Related
	13	Sales & Related
	14	Sales & Related
	15	Sales & Related
	16	Sales & Related
	17	Sales & Related
	18	Sales & Related
	19	Sales & Related
	20	Sales & Related
	21	Sales & Related
	22	Sales & Related
	23	Management
	24	Management
	25	Management
	26	Management

Python Code: `df[df['occupation'].isin(['Sales & Related', 'Management'])][['occupation']]`

occupation	
193	Sales & Related
194	Sales & Related
195	Sales & Related
196	Sales & Related
197	Sales & Related
...	...
12679	Sales & Related
12680	Sales & Related
12681	Sales & Related
12682	Sales & Related
12683	Sales & Related
1931 rows × 1 columns	